Phys 339 Final project (Chapter 12 Nonlinear Mechanics and Chaos)

Why this area of classical mechanics is important

What we have learned so far has worked well for linear systems. How ever when we leave out comfort zone of linear equations problems tend to arise. One of these problems that modeling some non-linear systems exhibits is chaos. This appears in all sorts of applications, fluid flow,population growth even the spread of disease. The behavior of chaotic systems is can be extremely complicated, thus new ways of interpreting these systems where created. Examples of these include state-space orbits, Poincare section, and bifurcation diagrams. There are some systems that are simple enough to still use traditional methods of interpretation and we will look at some of these in this paper.

Just to be clear not all non-linear systems are chaotic, But non-linearity is essential for chaos. This is important because Nonlinear Equations don't Obey the Superposition Principle. The super position principle states that any linear combination of solutions gives us another solution. For an example of this let us look at a relatively simple equation.

$$\ddot{x} = \epsilon(1 - x^2)\dot{x} - \alpha x \tag{1}$$

To find the equilibrium point we can set $\dot{x}$ and $\ddot{x}$ equal to zero and solve for x.

$$0 = \alpha x \tag{2}$$

From this we see that the only possible solution is Zero. We can gain more insight in to whether this equilibrium point is stable by piloting it in phase space. In order to graph in phase space we will have the x position on the horizontal Axis and $\dot{x}$ on the vertical axis. This requires solving the problem numerically using Euler's method. For our starting values where chosen to use $\alpha = 1$, $\epsilon = 8.53$, $\dot{x} = 0$, and $x = .1$. With an X value so close to the

equilibrium point we should see it tend toward the equilibrium point if it is stable or began to move away if the point is not stable.

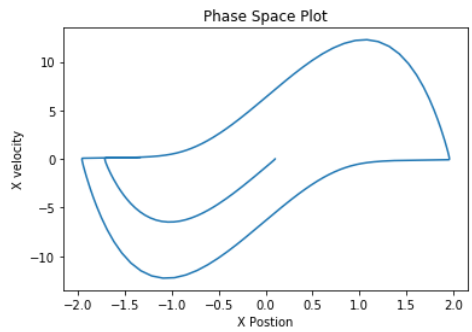


Figure 1: As we can see, this point is not a stable equilibrium point

As you can see form this the particle drifts away from the equilibrium point. This leads us too come to the conclusion that this is not a stable point. We can explore the movement more thoroughly thought the use of a quiver plot and additionally stream plot. To do this we will first create a grid mesh. This will create a list of X and $\dot{x}$ values that span the graph. Then, for each of these points we will calculate a vector representation of the change in x and $\dot{x}$. These vectors horizontal component are equal to $\dot{x}$ and its vertical component are equal to $\ddot{x}$.

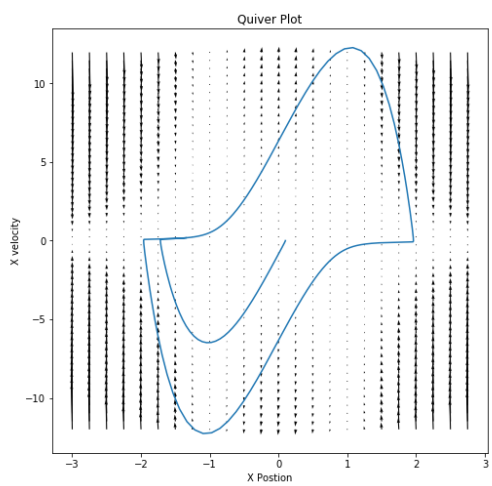
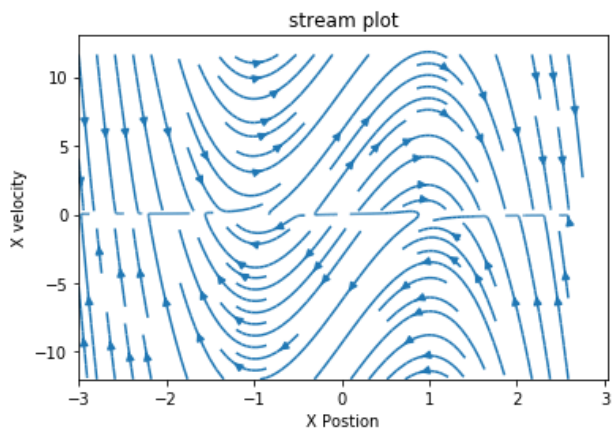
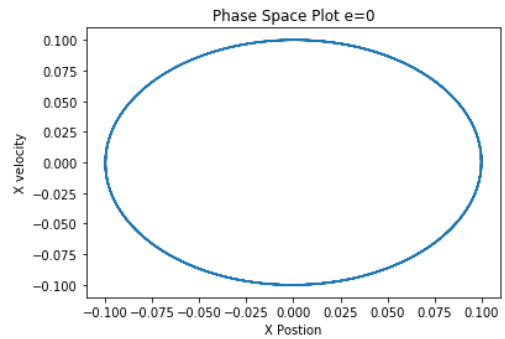


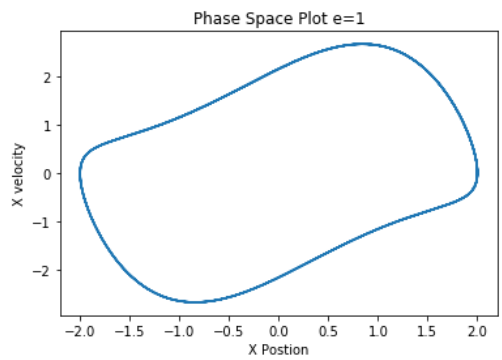
Figure 2: This is the Quiver Plot, it has plotted vectors is a grid pattern. The horizontal component representing the $\dot{x}$ and the vertical component representing $\ddot{x}$ at each placement.

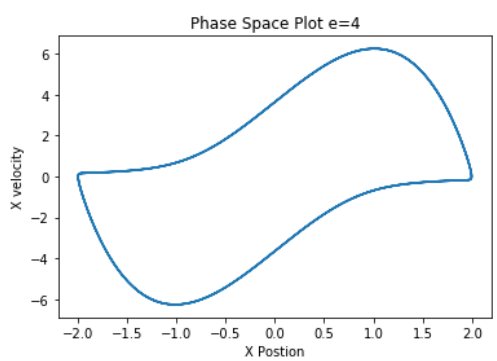
The quiver plot helps us make sense of the movement of the particle and path of the object. This is done better by the Stream plot . the Stream plot takes in the same data as the quiver plot and stitches it together to make a more coherent story.

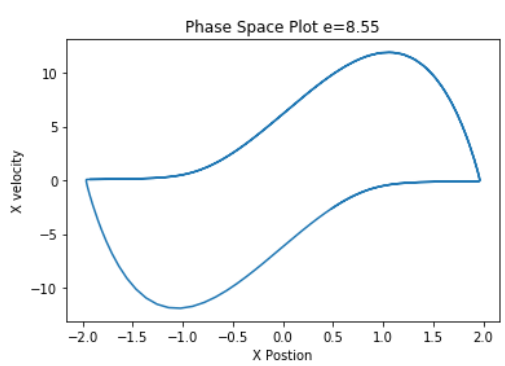


Although there is only one equilibrium point and that point is not stable, there is an equilibrium orbit also called a limit cycle, This orbit is stable as when we look at the stream plot we can put an atom anywhere on the chart and it will eventually return to the same orbit. This is one of the fundamental reasons the current equation for motion in not chaotic as we can place the atom in differing locations and if we wait long enough the position will be in a similar place no matter what. Where as in a chaotic system even the smallest difference in position will result in an exponential difference. To explore this phenomena further I went ahead and set $\epsilon = 0, 1, 4, 8.55$. I plotted each and only took the tail of the data to make sure it had reach its limit cycle. All of there graphs lack chaos as they find a steady orbit.


Phase Space Plot e=0
X velocity
X Postion


Phase Space Plot e=1
X velocity
X Postion


Phase Space Plot e=4
X velocity
X Postion


Phase Space Plot e=8.55
X velocity
X Postion

When comparing all the diagrams we see ϵ does effect the shape of the graph, but not the fact that it does eventually reach a limit cycle. Finally let Us add a driving force in to the equation.

$$\ddot{x} = \epsilon(1 - x^2)\dot{x} - \alpha x + F \cos \frac{2\pi t}{T} \tag{3}$$

For this We will be using Euler's method to solve this equation numerically. we will be solving this two times with different initial conditions. This will hopefully let us determine whether or not the current equation parameters will result a chaotic system or not. For the initial conditions used in both equations $F = 1.2$,$\alpha = 1$,$\epsilon = 8.53$, and $\dot{x}_1, \dot{x}_2 = 0$. We will be varying the starting position,$x_1 = 2$ and $x_2 = 2.05$. As you can see it is a very small difference.

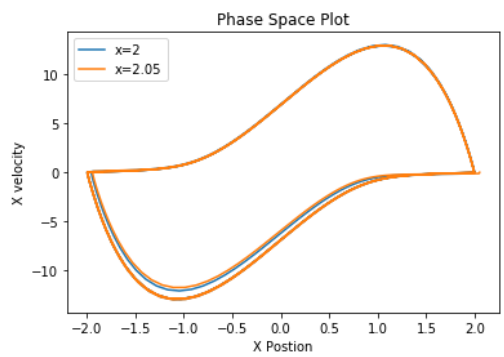


Figure 3: $F = 1.2$,$\alpha = 1$,$\epsilon = 8.53$, and $\dot{x}_1, \dot{x}_2 = 0$

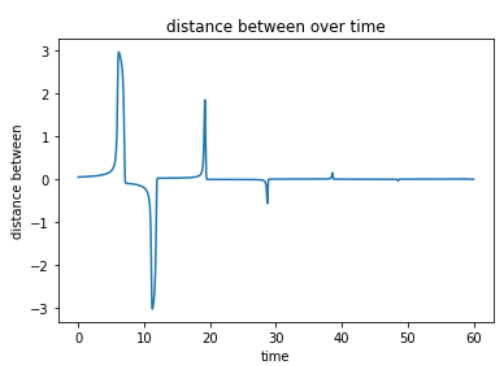


Figure 4: $F = 1.2$,$\alpha = 1$,$\epsilon = 8.53$, and $\dot{x}_1, \dot{x}_2 = 0$

Over all we see that although we have added the driver we do see the two values for the phase space converge over a long enough period of time. We also see in the phase space, two both settle in to a periodic motion. This leads us to conclude that this motion is not chaotic as the distance between them is not increasing exponentially and the motion in the phase space in periodic.

moving on to the second set of values. $F = 5$,$\alpha = 1$,$\epsilon = 3$,$\omega = 1.788$ and $\dot{x}_1, \dot{x}_2 = 0$. Note that these are the updated values yet to be added to the question and ω replaces the value of $\frac{2\pi}{T}$.

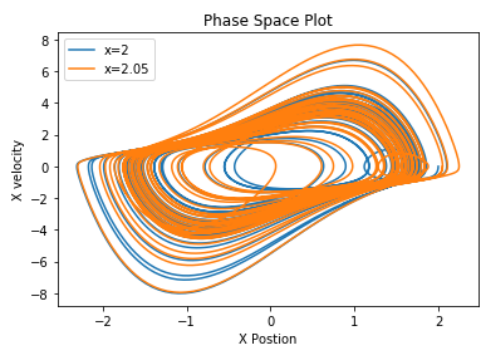


Figure 5: $F = 5$,$\alpha = 1$,$\epsilon = 3$,$\omega = 1.788$ and $\dot{x}_1, \dot{x}_2 = 0$

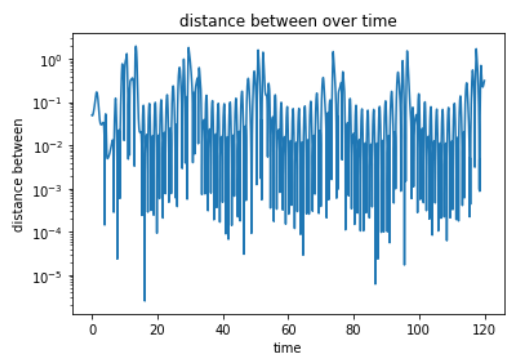


Figure 6: $F = 5$,$\alpha = 1$,$\epsilon = 3$,$\omega = 1.788$ and $\dot{x}_1, \dot{x}_2 = 0$

With these small change in variables we see that these chart look much different from the previous set. The first thing you notice is that the phase space plot is any thing but periodic. Over a total run time of 120 seconds it did not repeat its self. Further more if we look at the distance between the two starting points is phase space they don't seem to

be reducing or forming some sort of periodic structure. This lead us to conclude with this set of initial conditions this system is Chaotic. Although it is strange that the two starting points don't get farther apart in phase space.

That concludes the questions for this chapter we hope this was informative. To create such models we used Euler's method to solve numerically. Then, to graph is phase space you must graph x on the horizontal axis and $\dot{x}$ on the vertical. Good Luck.